Anusha Anand

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Experience

Citi Bank | **Tech Mahindra Americas** | *Senior Software Engineer* Feb 2025 – Present

- Architected distributed **batch and streaming ETL pipelines** in **PySpark** on **Apache Spark** and **Kafka**, processing **10M+ financial records daily** and cutting manual validation overhead **35%**.
- Built an AI-driven **data quality framework** with **anomaly detection** and rule-based validation across **5+ enterprise data workflows**, reducing data-defect escapes **28%**.
- Built end-to-end observability with **AWS CloudWatch** alerting, cutting **mean time to detect** from hours to **under 10 minutes**.
- Engineered a **Python** ML system mining historical failure logs to generate targeted tests, raising **QA coverage 40%** above baseline.
- Led **3+ concurrent projects** end to end; earned engineering excellence recognition and a **Star Award** nomination.

iHeartMedia | *Software Engineering Intern* Jun 2024 – Aug 2024

- Designed and deployed an **event-driven serverless** scheduler on **AWS Lambda, API Gateway, S3, SNS, and SQS**, defined as **infrastructure as code** in **AWS CDK**, automating **20+ Boomi workflows**.
- Secured **8+ API Gateway endpoints** with custom **Lambda authorizers** and **JWT**-based access control.
- Built **React.js** operational dashboards with advanced filtering and **6 real-time metrics** across **5,000+ records**.

Nomura Research Institute Financial Technologies | *Associate Software Engineer* Aug 2020 – Jul 2022

- Developed and maintained **Java batch jobs** on **Spring Framework** and **Oracle** databases, writing **SQL** for data processing and a grid-based query UI in **HTML/JavaScript**.
- Automated batch execution with **Shell scripting** and **Jenkins** pipelines; owned end-to-end delivery through deployment.
- Led client-facing **Java** reporting projects, mentored juniors, and ran **code reviews**; chosen for **onsite client work in Japan**.

Nomura Research Institute Financial Technologies | *Software Engineer Intern* Jan 2020 – Mar 2020

- Created **50+ automated test cases** in **Katalon Studio**, building **Java/Groovy** test automation with QA and dev teams.

Projects

Resilient Transaction Management System *Python | Sockets | Distributed Systems*

- Developed a fault-tolerant **distributed transaction system** in **Python** using the **two-phase commit (2PC)** protocol over **socket communication**. Ensured **data consistency** and reliability across nodes during **network failures**.

YouTube Video Query Assistant *LangChain | OpenAI Embeddings | Vector DB | Streamlit*

- Built a **LangChain** question-answering system over YouTube videos, processing transcripts with **OpenAI Embeddings** stored in a **vector database**. Enhanced querying with recursive search and error handling behind a **Streamlit** interface.

DishLens: AI Recipe Finder *React | FastAPI | Gemini 2.0 Flash | Docker*

- Shipped a zero-login food recognition app covering **101 dish types**, classifying images with **Gemini 2.0 Flash** and a **CLIP** fallback. Deployed a **FastAPI** backend in **Docker** on Hugging Face Spaces, with a **React/Vite** frontend surfacing macros and recipes.

Education

The University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2022 – May 2024

Sikkim Manipal Institute of Technology | *B.Tech Computer Science and Engineering, CGPA 9.25/10* 2016 – 2020

Skills

Languages: Python, SQL, Java, JavaScript, Shell, HTML/CSS

Frameworks: Spring, FastAPI, React.js, LangChain, OpenAI Embeddings, Vector Databases, JWT

Data Engineering: Apache Spark, PySpark, Kafka, ETL Pipelines, Batch & Streaming, Data Quality, Distributed Systems, Oracle DB

AWS & Cloud: AWS (Lambda, API Gateway, S3, SNS, SQS, CloudWatch, CDK), Docker, Jenkins, CI/CD

Certifications

Databricks Certified Data Engineer Associate (Udemy) Oct 2026
AWS Certified Machine Learning Engineer, Associate Oct 2025
AWS Certified Solutions Architect, Associate Dec 2024
Boomi Associate Integration Developer Aug 2024